

```
# Introducing Pandas
```

```
!pip install pandas
```

```
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (2.2.2)
Requirement already satisfied: numpy>=1.26.0 in /usr/local/lib/python3.12/dist-packages (from pandas) (2.0.2)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas) (2026.2)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas) (1.
```

```
!pip install numpy
```

```
Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages (2.0.2)
```

```
import numpy as np
import pandas as pd
```

```
data = ["Revanth","Ritesh","Srivardhan","Madhur"]
mydata = pd.Series(data)
mydata
```

```

      0
0    Revanth
1     Ritesh
2  Srivardhan
3     Madhur
```

dtype: object

```
data = ["Revanth","Ritesh","Srivardhan","Madhur"]
mydata = pd.Series(data, index=[10,11,12,13])
mydata
```

```

      0
10   Revanth
11    Ritesh
12  Srivardhan
13    Madhur
```

dtype: object

```
data = ["Revanth","Ritesh","Srivardhan","Madhur"]
mydata = pd.Series(data, index=["a","b","c","d"])
mydata
```

```

      0
a    Revanth
b     Ritesh
c  Srivardhan
d     Madhur
```

dtype: object

```
mydata = pd.Series("10", index = list(range(0,6)))
mydata
```

0
0 10
1 10
2 10
3 10
4 10
5 10

dtype: object

data = {"Name" : ["A","B","C"], "Age" : [21,23,15], "Education" : ["BBA","MS","Btech"]} }

dt = pd.DataFrame(data)
dt

	Name	Age	Education
0	A	21	BBA
1	B	23	MS
2	C	15	Btech

df = pd.read_csv("/content/German-Credit_1.csv")

df.head()

	OBS	DURATION	NEW_CAR	USED_CAR	FURNITURE	RADIO_TV	EDUCATION	RETRAINING	AMOUNT	INSTALL_RATE	...	PROP_UNKN_NONE	
0	1	6.0	0.0	0.0	0.0	1.0	0.0	0.0	1169.0	4.0	...	0.0	6
1	2	48.0	0.0	0.0	0.0	1.0	0.0	0.0	5951.0	2.0	...	0.0	2
2	3	12.0	0.0	0.0	0.0	0.0	1.0	0.0	2096.0	2.0	...	0.0	4
3	4	42.0	0.0	0.0	1.0	0.0	0.0	0.0	7882.0	2.0	...	0.0	4
4	5	24.0	1.0	0.0	0.0	0.0	0.0	0.0	4870.0	3.0	...	1.0	5

5 rows × 23 columns

df.tail()

	OBS	DURATION	NEW_CAR	USED_CAR	FURNITURE	RADIO_TV	EDUCATION	RETRAINING	AMOUNT	INSTALL_RATE	...	PROP_UNKN_NONE	
995	996	12.0	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	...	NaN	
996	997	30.0	0.0	1.0	0.0	0.0	0.0	0.0	3857.0	4.0	...	0.0	
997	998	12.0	0.0	0.0	0.0	1.0	0.0	0.0	804.0	4.0	...	0.0	
998	999	45.0	0.0	0.0	0.0	1.0	0.0	0.0	1845.0	4.0	...	1.0	
999	1000	45.0	0.0	1.0	0.0	0.0	0.0	0.0	NaN	3.0	...	0.0	

5 rows × 23 columns

df.shape

(1000, 23)

df.columns

Index(['OBS', 'DURATION', 'NEW_CAR', 'USED_CAR', 'FURNITURE', 'RADIO_TV',
'EDUCATION', 'RETRAINING', 'AMOUNT', 'INSTALL_RATE', 'CO_APPLICANT',
'GUARANTOR', 'REAL_ESTATE', 'PROP_UNKN_NONE', 'AGE', 'OTHER_INSTALL',
'RENT', 'OWN_RES', 'NUM_CREDITS', 'NUM_DEPENDENTS', 'TELEPHONE',
'FOREIGN', 'RESPONSE'],
dtype='object')

df1 = df["INSTALL_RATE"]

df1.head()

INSTALL_RATE	
0	4.0
1	2.0
2	2.0
3	2.0
4	3.0

dtype: float64

df.info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 23 columns):
#   Column                Non-Null Count  Dtype
---  -
0   OBS                    1000 non-null  int64
1   DURATION               998 non-null   float64
2   NEW_CAR                999 non-null   float64
3   USED_CAR               999 non-null   float64
4   FURNITURE              999 non-null   float64
5   RADIO_TV               997 non-null   float64
6   EDUCATION              996 non-null   float64
7   RETRAINING             996 non-null   float64
8   AMOUNT                 990 non-null   float64
9   INSTALL_RATE           993 non-null   float64
10  CO_APPLICANT           992 non-null   float64
11  GUARANTOR              993 non-null   float64
12  REAL_ESTATE            993 non-null   float64
13  PROP_UNKN_NONE         993 non-null   float64
14  AGE                    990 non-null   float64
15  OTHER_INSTALL          991 non-null   float64
16  RENT                   992 non-null   float64
17  OWN_RES                993 non-null   float64
18  NUM_CREDITS            994 non-null   float64
19  NUM_DEPENDENTS         993 non-null   float64
20  TELEPHONE              995 non-null   float64
21  FOREIGN                994 non-null   float64
22  RESPONSE               997 non-null   float64
dtypes: float64(22), int64(1)
memory usage: 179.8 KB
```

df.isnull().sum()

	0
OBS	0
DURATION	2
NEW_CAR	1
USED_CAR	1
FURNITURE	1
RADIO_TV	3
EDUCATION	4
RETRAINING	4
AMOUNT	10
INSTALL_RATE	7
CO_APPLICANT	8
GUARANTOR	7
REAL_ESTATE	7
PROP_UNKN_NONE	7
AGE	10
OTHER_INSTALL	9
RENT	8
OWN_RES	7
NUM_CREDITS	6
NUM_DEPENDENTS	7
TELEPHONE	5
FOREIGN	6
RESPONSE	3

dtype: int64

`df.mean()`

0	
OBS	500.500000
DURATION	20.917836
NEW_CAR	0.234234
USED_CAR	0.103103
FURNITURE	0.180180
RADIO_TV	0.279840
EDUCATION	0.050201
RETRAINING	0.097390
AMOUNT	3273.168687
INSTALL_RATE	2.972810
CO_APPLICANT	0.041331
GUARANTOR	0.052367
REAL_ESTATE	0.281974
PROP_UNKN_NONE	0.155086
AGE	35.505051
OTHER_INSTALL	0.186680
RENT	0.178427
OWN_RES	0.713998
NUM_CREDITS	1.406439
NUM_DEPENDENTS	1.156093
TELEPHONE	0.403015
FOREIGN	0.037223
RESPONSE	0.700100

dtype: float64

df.min()

	0
OBS	1.0
DURATION	4.0
NEW_CAR	0.0
USED_CAR	0.0
FURNITURE	0.0
RADIO_TV	0.0
EDUCATION	0.0
RETRAINING	0.0
AMOUNT	250.0
INSTALL_RATE	1.0
CO_APPLICANT	0.0
GUARANTOR	0.0
REAL_ESTATE	0.0
PROP_UNKN_NONE	0.0
AGE	19.0
OTHER_INSTALL	0.0
RENT	0.0
OWN_RES	0.0
NUM_CREDITS	1.0
NUM_DEPENDENTS	1.0
TELEPHONE	0.0
FOREIGN	0.0
RESPONSE	0.0

dtype: float64

```
print(df['AGE'].mean())
```

35.505050505050505

```
df.std()
```

```

0
OBS      288.819436
DURATION 12.064656
NEW_CAR   0.423731
USED_CAR  0.304246
FURNITURE 0.384530
RADIO_TV  0.449146
EDUCATION 0.218469
RETRAINING 0.296636
AMOUNT    2826.183525
INSTALL_RATE 1.119055
CO_APPLICANT 0.199154
GUARANTOR 0.222877
REAL_ESTATE 0.450187
PROP_UNKN_NONE 0.362169
AGE       11.342694
OTHER_INSTALL 0.389851
RENT      0.383065
OWN_RES   0.452118
NUM_CREDITS 0.576302
NUM_DEPENDENTS 0.363126
TELEPHONE 0.490750
FOREIGN   0.189404
RESPONSE  0.458444

```

```
dtype: float64
```

```
df.describe()
```

	OBS	DURATION	NEW_CAR	USED_CAR	FURNITURE	RADIO_TV	EDUCATION	RETRAINING	AMOUNT	INSTALL_RA
count	1000.000000	998.000000	999.000000	999.000000	999.000000	997.000000	996.000000	996.000000	990.000000	993.0000
mean	500.500000	20.917836	0.234234	0.103103	0.18018	0.279840	0.050201	0.097390	3273.168687	2.9728
std	288.819436	12.064656	0.423731	0.304246	0.38453	0.449146	0.218469	0.296636	2826.183525	1.1190
min	1.000000	4.000000	0.000000	0.000000	0.00000	0.000000	0.000000	0.000000	250.000000	1.0000
25%	250.750000	12.000000	0.000000	0.000000	0.00000	0.000000	0.000000	0.000000	1371.250000	2.0000
50%	500.500000	18.000000	0.000000	0.000000	0.00000	0.000000	0.000000	0.000000	2321.500000	3.0000
75%	750.250000	24.000000	0.000000	0.000000	0.00000	1.000000	0.000000	0.000000	3970.500000	4.0000
max	1000.000000	72.000000	1.000000	1.000000	1.00000	1.000000	1.000000	1.000000	18424.000000	4.0000

```
8 rows × 23 columns
```

```
df2 = (df['AMOUNT'] > 1000)
```

```
print(df2.head())
```

```

0    True
1    True
2    True
3    True
4    True
Name: AMOUNT, dtype: bool

```

```
df2
```

AMOUNT

0	True
1	True
2	True
3	True
4	True
...	...
995	False
996	True
997	False
998	True
999	False

LOC, ILOC

```
df2 = df.iloc[:, 1:5]  
df2.head()
```

	DURATION	NEW_CAR	USED_CAR	FURNITURE
0	6.0	0.0	0.0	0.0
1	48.0	0.0	0.0	0.0
2	12.0	0.0	0.0	0.0
3	42.0	0.0	0.0	1.0
4	24.0	1.0	0.0	0.0

```
df2 = df.iloc[1:10, 1:5]  
df2.head()  
print(df2.shape)
```